Same Mean, Different Deadline Risk

AP Statistics · Unit 2 · Topic 2.9 · B: 2026-11-02 · E: 2026-11-23 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU VAR-5** — Probability distributions may be used to model variation in populations.
- **LO VAR-5.C** — Calculate parameters for a discrete random variable. [Skill 3.B]
- **EK VAR-5.C.1** — A numerical value measuring a characteristic of a population or the distribution of a random variable is known as a parameter, which is a single, fixed value.
- **EK VAR-5.C.2** — The mean, or expected value, for a discrete random variable X is $\mu_X = \sum x_i \cdot \mathrm{P}(x_i)$.
- **EK VAR-5.C.3** — The standard deviation for a discrete random variable X is $\sigma_X = \sqrt{[\sum (x_i - \mu_X)^2 \cdot \mathrm{P}(x_i)]}$.
- **LO VAR-5.D** — Interpret parameters for a discrete random variable. [Skill 4.B]
- **EK VAR-5.D.1** — Parameters for a discrete random variable should be interpreted using appropriate units and within the context of a specific population.

Essential question: When two probability models have the same mean, how can standard deviation change the decision?

DOK-3 skill: 4.B · **FRQ pattern:** `same-mean-different-sd-which-parameter-answers-the-question`

DOK rationale: Part (c) weighs equal means against sharply different variability for a deadline-sensitive decision, carries both parameters through a transformation, and explains the practical consequence hidden by a mean-only comparison.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask for a route choice before discussing parameter formulas.
- Begin the parameter portion of the combined 4.7–4.8 follow-along at minute 5.
- Return to the route models at minute 33. Circulate and connect spread to this customer's deadline.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose a route and cite one model feature.
- Complete the video follow-along about means, standard deviations, and their interpretations.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What do the equal means say, and what do they leave unresolved?
- Which route places probability beyond the 40-minute deadline?
- When every outcome increases by five, what happens to each distance from its new mean?

- Would your decision be the same for a customer who valued the chance of arriving extremely early?

Adult role
Solo teacher. Circulate 5 pairs. Require both a parameter and a deadline-specific display feature; do not let same average stand in for same reliability.

[WARN] WATCH FOR

- Interpreting a standard deviation as the exact deviation of every delivery.
- Using sample standard deviation with denominator 7 instead of the probability-model denominator 8.
- Adding five to standard deviation when loading time is constant.

SCORING PART (c) — E / P / I

Expected elements

- **chooses-low-variability-route** (required) — Chooses Route A and cites its 3-minute standard deviation versus Route B's 12-minute standard deviation
- **connects-display-to-deadline** (required) — Uses Route A's 25-to-35 range and no outcomes above 40 versus Route B's two outcomes above 40
- **diagnoses-mean-only-claim** (required) — Explains that equal centers do not imply equal consistency or equal late risk
- **transforms-both-parameters** (required) — Adds 5 to both means, leaves both standard deviations unchanged, and reevaluates the decision

E	Chooses Route A using both standard deviation and deadline evidence, explains the mean-only misconception, correctly transforms all four parameters, and confirms the post-shift decision.
P	Chooses Route A with relevant variability evidence but omits the practical mean-only consequence, one transformed parameter, or the revised deadline comparison.
I	Says the routes are interchangeable because their means match, adds 5 to either standard deviation, or chooses without connecting variability to lateness.

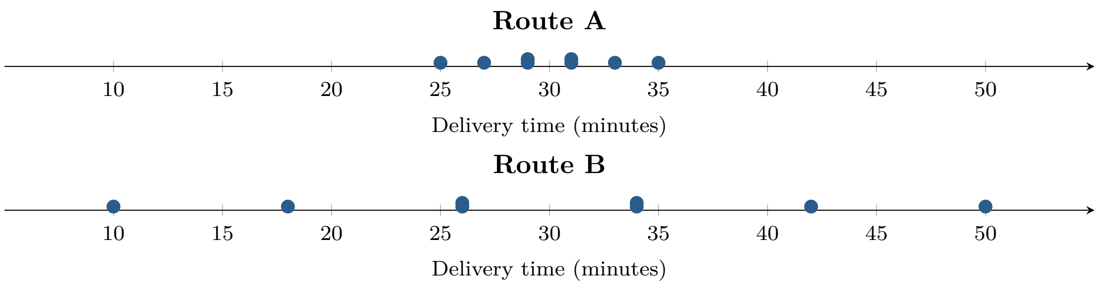
Common mistakes

- Treating standard deviation as extra minutes added to every delivery
- Claiming every Route B delivery is 12 minutes from 30
- Adding 5 to the standard deviations when loading time is constant
- Using the equal means alone to conclude equal probabilities of missing 40 minutes

[STAR] TODAY'S PROBLEM — annotated

Suppose a dispatcher models delivery time using eight equally likely outcomes for each route, shown in the dotplots. An analyst reports that both routes have mean 30 minutes, with $\sigma_A = 3$ minutes and $\sigma_B = 12$ minutes.

The dispatcher says, *The routes have the same mean, so it does not matter which one a customer gets.* A customer has 40 minutes before an appointment and places a high cost on being late.



FIRST TAKE (5 min)

Which route would you assign to this customer? Cite one number or dotplot feature.
Key: Not graded. Route B's very fast outcomes may attract some students before deadline risk is made explicit.

Rules callout "MEAN, STANDARD DEVIATION, AND SHIFTS (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** For each route, state its minimum and maximum modeled time and count the outcomes longer than 40 minutes.

Key: Route A has minimum 25, maximum 35, and 0/8 outcomes above 40 minutes. Route B has minimum 10, maximum 50, and $2/8 = 25\%$ above 40: 42 and 50.

DOK 2 **(b)** Verify the reported mean and population standard deviation for each route from the eight equally likely outcomes. Show the sums of squared deviations.

Key: Both sets sum to 240, so each mean is $240/8 = 30$ minutes. For Route A, the deviations are $-5, -3, -1, -1, 1, 1, 3, 5$ and the squared deviations sum to $25 + 9 + 1 + 1 + 1 + 1 + 9 + 25 = 72$, so $\sigma_A = \sqrt{72/8} = 3$. For Route B, deviations are $-20, -12, -4, -4, 4, 4, 12, 20$ and their squares sum to $400 + 144 + 16 + 16 + 16 + 16 + 144 + 400 = 1152$, so $\sigma_B = \sqrt{1152/8} = 12$ minutes.

DOK 3 **(c)** Decide which route better matches this customer's priority. Cite the parameter and dotplot feature that settle your choice, and explain what the dispatcher's mean-only argument would mislead a reader into believing. Then suppose every delivery gets 5 minutes of loading time: state both transformed means and standard deviations and decide whether your route choice changes.

Key: Choose Route A: $\sigma_A = 3$ rather than $\sigma_B = 12$, and none of its times exceed 40, versus $2/8 = 25\%$ for Route B. Equal means describe center only, so the dispatcher's claim conceals unequal reliability and deadline risk. Adding 5 makes both means 35 but leaves the standard deviations 3 and 12 because a constant shift preserves deviations. The new ranges are 30–40 and 15–55, so Route A remains the supported choice.

EXIT

One thing the video changed about my first take: ____________________